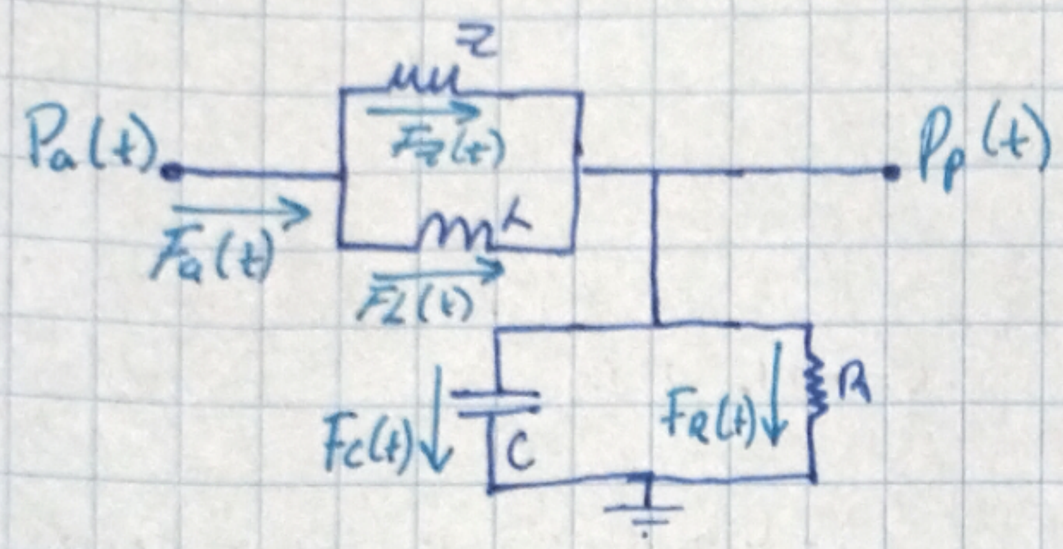
Práctica 2: Sistema Cardiovascular.

5-oct-2026.



- Modelado y control del sistema cardiovascular mediante el modelo Windkessel.
- Análisis nodal del circuito.

→ Ecuaciones principales. $F_a(t) = F_Z(t) + F_L(t) = F_C(t) + F_R(t)$.

- $F_Z(t) = \dfrac{P_a(t) - P_p(t)}{Z}$
- $F_L(t) = \dfrac{1}{L}\int [P_a(t) - P_p(t)]\,dt$
- $F_C(t) = C \cdot \dfrac{dP_p(t)}{dt}$
- $F_R(t) = \dfrac{P_p(t)}{R}$

→ Transformada de Laplace. $F_a(s) = F_Z(s) + F_L(s) = F_C(s) + F_R(s)$.

- $F_Z(s) = \dfrac{P_a(s) - P_p(s)}{Z}$
- $F_L(s) = \dfrac{P_a(s) - P_p(s)}{Ls}$
- $F_C(s) = Cs \cdot P_p(s)$
- $F_R(s) = \dfrac{P_p(s)}{R}$

→ Procedimiento Algebraico. $F_Z(s) + F_L(s) = F_C(s) + F_R(s)$

$$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = Cs \cdot P_p(s) + \frac{P_p(s)}{R}$$

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s)}{Ls} - \frac{P_p(s)}{Ls} = Cs \cdot P_p(s) + \frac{P_p(s)}{R}$$

$$P_a(s)\left(\frac{1}{Z} + \frac{1}{Ls}\right) = P_p(s)\left(Cs + \frac{1}{R} + \frac{1}{Ls} + \frac{1}{Z}\right)$$

$$P_a(s)\left(\frac{Ls + Z}{LZs}\right) = P_p(s)\left(\frac{CLRZs^2 + LZs + RZ + RLs}{RLZs}\right)$$

$$P_a(s)\left(\frac{Ls + Z}{LZs}\right) = P_p(s)\left(\frac{CLRZs^2 + (Lz + RL)s + RZ}{RLZs}\right)$$

$$P_a(s)\left(\frac{Ls + Z}{LZs}\right) = P_p(s)\left(\frac{CLRZs^2 + L(Z + R)s + RZ}{RLZs}\right)$$

$$\frac{P_p(s)}{P_a(s)} = \frac{\frac{Ls + Z}{LZs}}{\frac{CLRZs^2 + L(Z+R)s + RZ}{RLZs}}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + RZ}{CLRZs^2 + L(Z+R)s + RZ}$$

→ Estabilidad del Sistema en Lazo Abierto.

→ Polos del Sistema:

$$CLRZs^2 + L(Z+R)s + RZ = \emptyset$$

$$a = CLRZ \qquad b = L(Z+R) \qquad c = RZ$$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-[L(Z+R)] \pm \sqrt{[L(Z+R)]^2 - 4(CLRZ)(RZ)}}{2(CLRZ)}$$

$$= \frac{-L(Z+R) \pm \sqrt{L^2(Z+R)^2 - 4CLR^2Z^2}}{2(CLRZ)}$$

→ Hipotenso: $Z = 0.010$, $L = 0.001$, $R = 1.000$, $C = 3.500$

$$\lambda_{1,2} = \frac{-L(Z+R) \pm \sqrt{L^2(Z+R)^2 - 4CLR^2Z^2}}{2(CLRZ)}$$

$$= \frac{-(0.001)(0.01+1) \pm \sqrt{(0.001)^2(0.01+1)^2 - 4(3.5)(0.001)(1)^2(0.01)^2}}{2 \cdot (3.5)(0.001)(1)(0.01)}$$

$$= \frac{-0.00101 \pm \sqrt{(1.02010 \times 10^{-6}) - (1.4 \times 10^{-6})}}{7 \times 10^{-5}} = \frac{-0.00101 \pm \sqrt{-3.799 \times 10^{-7}}}{7 \times 10^{-5}}$$

$$= \frac{-0.00101 \pm (-3.799 \times 10^{-7} i)}{7 \times 10^{-5}} \qquad \lambda_1 = -14.4286 + 8.8051\,i \qquad \lambda_2 = -14.4286 - 8.8051\,i$$

∴ El sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario de entrada.

→ Normotenso: $Z = 0.010$, $L = 0.001$, $R = 1.000$, $C = 1.500$

$$\lambda_{1,2} = \frac{-(0.001)(0.01+1) \pm \sqrt{(0.001)^2(0.01+1)^2 - 4(1.5)(0.001)(1)^2(0.01)^2}}{2(1.5)(0.001)(1)(0.01)}$$

$$= \frac{-0.00101 \pm \sqrt{(1.02010 \times 10^{-6}) - (6 \times 10^{-7})}}{3 \times 10^{-5}} = \frac{-0.00101 \pm \sqrt{4.20100 \times 10^{-7}}}{3 \times 10^{-5}}$$

$$= \frac{-0.00101 \pm (0.00065)}{3 \times 10^{-5}} \qquad \lambda_1 = -12.0616 \qquad \lambda_2 = -55.2717$$

∴ El sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario de entrada.

→ Hipertenso: $Z=0.010$, $L=0.001$, $R=1.000$, $C=1.000$

$$\lambda_{1,2}=\frac{-(0.001)(0.01+1)\pm\sqrt{(0.001)^2(0.01+1)^2-4(1)(0.001)(1)^2(0.01)^2}}{2(1)(0.001)(1)(0.01)}$$

$$=\frac{-0.00101\pm\sqrt{(1.02010\times10^{-6})-(4\times10^{-7})}}{2\times10^{-5}}=\frac{-0.00101\pm\sqrt{6.20100\times10^{-7}}}{2\times10^{-5}}$$

$$=\frac{-0.00101\pm 7.8746\times10^{-4}}{2\times10^{-5}}$$

$\lambda_1=-11.1268$ $\lambda_2=-89.8732$

∴ El sistema presenta una respuesta estable y sobreamortiguada a un escalón unitario de entrada.

→ Error en estado estacionario. (Siempre a escalón unitario)

$$e(s)=\lim_{s\to\emptyset} sP_a(s)\left(1-\frac{P_p(s)}{P_a(s)}\right)=\lim_{s\to\emptyset} s\cdot\frac{1}{s}\left(1-\frac{LRs+RZ}{CLRZs^2+L(Z+R)s+RZ}\right)$$

$e(s)=\emptyset$ ∴ $e(t)=\emptyset V$

→ Modelo de Ecuaciones Integro-diferenciales.

$F_Z(t)+F_L(t)=F_C(t)+F_R(t)$

$$\frac{P_a(t)-P_p(t)}{Z}+\frac{1}{L}\int[P_a(t)-P_p(t)]dt=C\cdot\frac{dP_p(t)}{dt}+\frac{P_p(t)}{R}$$

Nota: No se pueden despejar funciones que se están derivando o integrando.

$$\frac{P_a(t)}{Z}+\frac{1}{L}\int[P_a(t)-P_p(t)]dt-C\cdot\frac{dP_p(t)}{dt}=\frac{P_p(t)}{R}+\frac{P_p(t)}{Z}$$

$$\left(\frac{1}{R}+\frac{1}{Z}\right)P_p(t)=\frac{P_a(t)}{Z}+\frac{1}{L}\int[P_a(t)-P_p(t)]dt-C\cdot\frac{dP_p(t)}{dt}$$

$$P_p(t)=\left[\frac{P_a(t)}{Z}+\frac{1}{L}\int[P_a(t)-P_p(t)]dt-C\cdot\frac{dP_p(t)}{dt}\right]\frac{RZ}{Z+R}$$

∴ Modelo